

SUSTAINABLE URBAN COMMUNITIES

Waterbox, Autonomous urban water Utilities

By **FLATBOX**

LOCATION

Guatemala

TEAM LEADER

Edwin Kestler

...

SOLUTION

Our solution:

Waterbox is a 2G IoT, AI managed autonomous Water Meter, it acquired, classifies, maps and compares water usage demand and maintain water utility services, extending services coverage and oversight.



Our pitch:



IoT IWC Awards 2017



The problem:

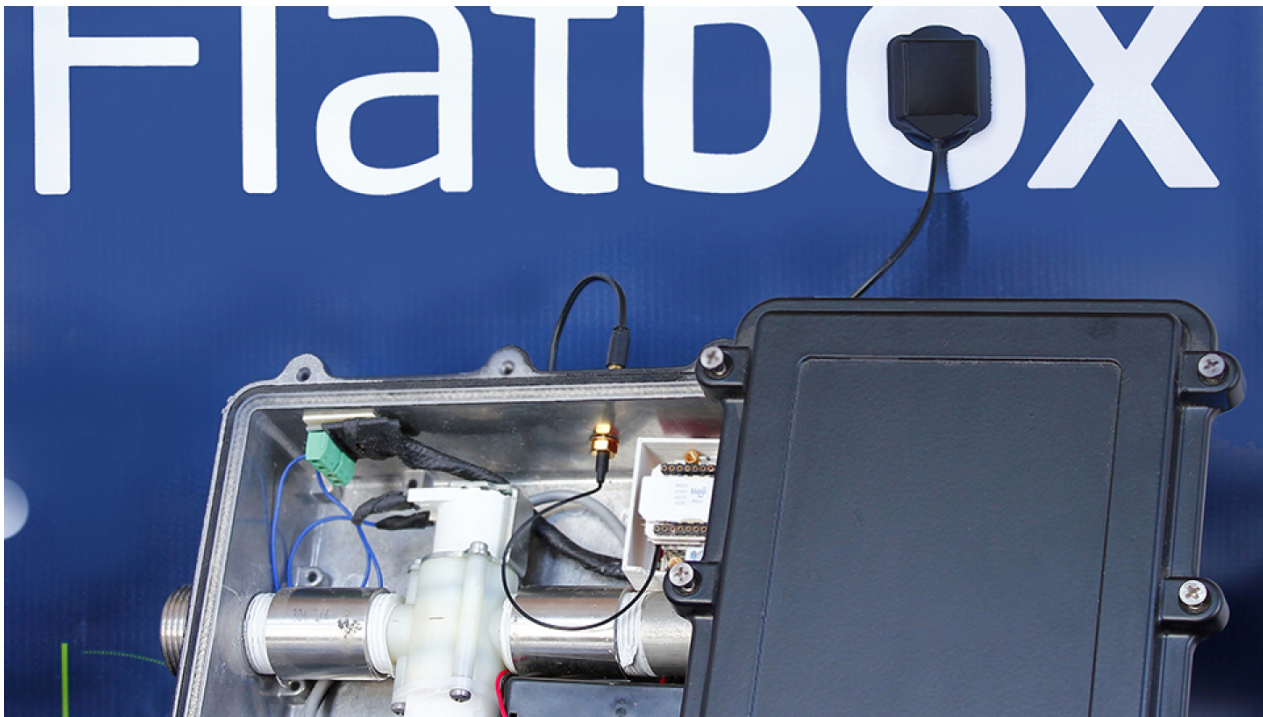


it significantly reduces the enormous Costs of providing water utility services; AI handles data acquired, classifies, maps and compares water usage demand, assigns billing profiles, uses machine learning to maintain the

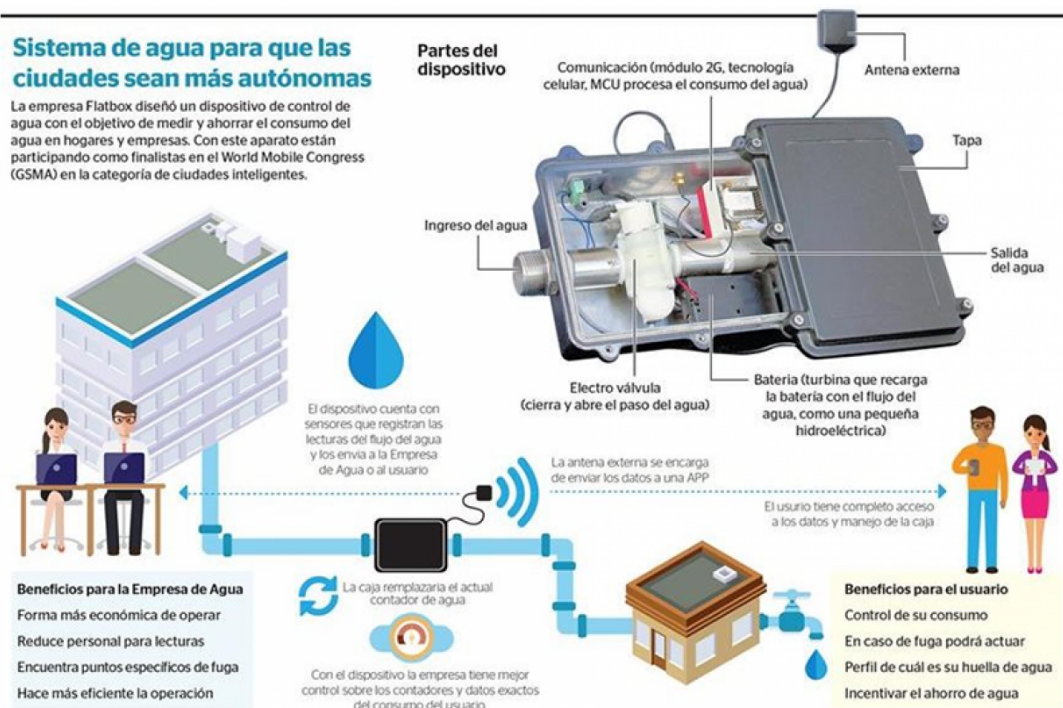
service, autonomously analyzing sensor data a self-recharging mechanism ,pressure, tamper and other sensors prevent water fraud and report potential problems such as leaks in the network without human supervision and its cost;



Designed to meet demands for House hold, Small and Medium Business Utilities it enables new pricing models and payment (SMS,Mobile, POS) to be introduced, such as PAYG, prepaid, and other for low income or remote rural areas.



Why our solution will solve the problem:



Bringing transparency, control and flexibility of consumption to utility services increased increased 15% of reliability of utility services and

Our solution's stage of development:



Research

Target Outcomes

Our target outcomes:



Better water management and monitoring results in a better usage of the resource service preventing scarcity and misuse, providing transparency and efficiency in the utility service, deployment will be done by user adoption (purchase) of the devices driven by reduced water fees incentives via reduction on first water usage data.

How we will measure our progress:



Outcome: Reduction in resource miss usage

Measurement Plan: reduced fee for low than average consumption measurements

Outcome: extended water coverage

Measurement Plan: implementation of PAYG and prepaid plans via SMS and mobile payments

Outcome: increased awerness in resource usage

Measurement Plan: logging website visits to water usage map provided to the user and dispatch of SMS threshold warning alerts.

The populations we will benefit initially:



Upper middle income economies (between \$3976 and \$12275 GNI)

Lower middle income economies (between \$1006 and \$3975 GNI)

Urban

Rural

Suburban

The regions we will benefit initially:



Europe and Central Asia

Latin America and the Caribbean

US and Canada

The countries we will benefit initially:



Colombia

El Salvador

Guatemala

Mexico

United States

Technology

The technologies we employ:



Consumer-facing software (mobile applications, cloud services)

Digital systems (machine learning, control systems, big data)

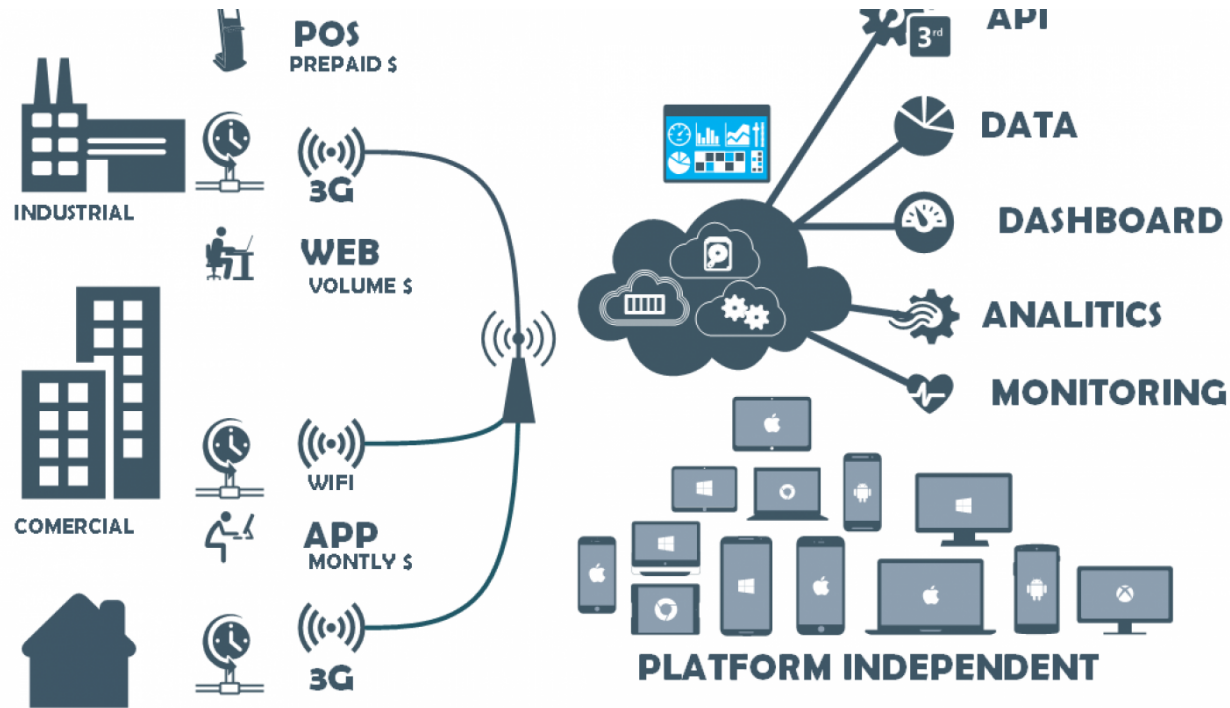
Electrical engineering

Mechanical engineering and hardware

Robotics



Why our solution is unique:



it uses energy harvesting by means of providing its own recharging energy from the water flow it measures

it uses AI to require minimum human supervision eliminating the high cost of monitoring centers and personnel

it uses Machine Learning to minimize the cost of system maintenance

it engages user awareness via notifications and alarms to directly involve end users in conservation and better usage ACTIONS.

Why our solution is human-centered:





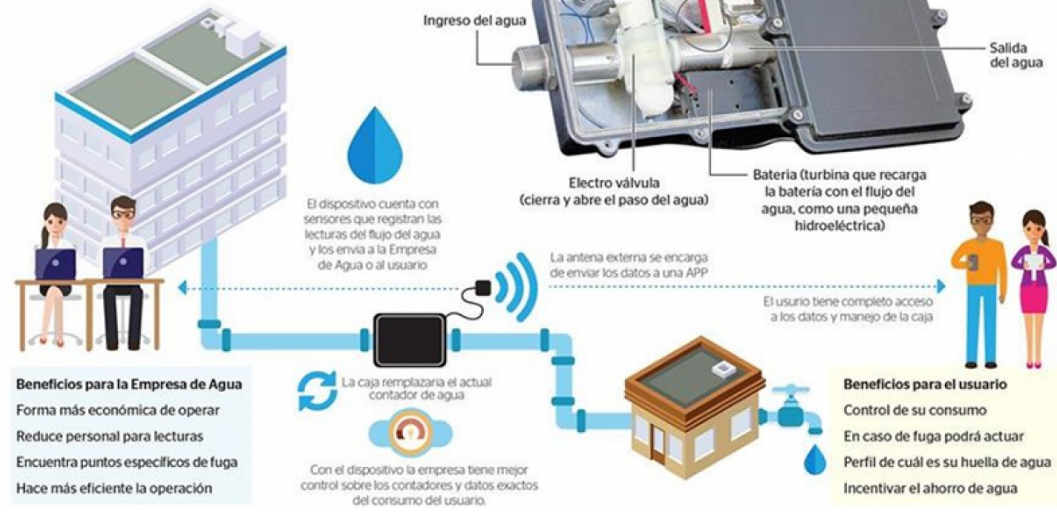
It enhances conservation actions by involving people in acknowledging their water usage, and providing them with tools to better manage (via thresholds) their resources, it extends the coverage of services by giving communities a simple way to implement an billing platform with out the need of information systems knowledge, right out the box enabling the implementation of water utility services that can be managed by communities right out of the box making such en devours self sustained from day one.

How people will access our solution:



Sistema de agua para que las ciudades sean más autónomas

La empresa Flatbox diseñó un dispositivo de control de agua con el objetivo de medir y ahorrar el consumo del agua en hogares y empresas. Con este aparato están participando como finalistas en el World Mobile Congress (GSMA) en la categoría de ciudades inteligentes.



it will be deployed via the site, providing full implementation kits (water meters) and by making the product opensource, enabling communities to build their own versions of the hardware water meter and uploading the firmware and providing a simple API to connect thir DIY device to the billing platform and registering users by simply enrolling their mobile phone numbers and the IMEI of the waterbox that provides the service.

Technology-Readiness Level:

6-8 (Demonstration)

[Our website](#) 

Business Plan



Our organization:

For-Profit

Where we are located:

Guatemala



How we will sustain our team financially:



so far we have been granted angel investor seed funds and a funds from a nonrefundable price from a local NGO; upon completion of first pilot program we plan on seeking funding from investors to expand the services in urban areas of Guatemala (private gated house compounds) about 1 Million house holds in Guatemala and from there make the project self sufficient to expand to rural areas, and seeking VC funds for international growth (about 18 month plan)

The factors limiting our success:



the actual make of the device, as any other IoT project the actual front cost of making the devices is our major growth limit, as increasing numbers changes costs and time tables for delivery of the devices, making, shipping and delivering the devices is our biggest logistics challenge.

How long we have been working on our solution:



1 year

How long it will take to develop a pilot:



3-6 months

How long it will take to scale beyond our pilot:



6-12 months

Our expected annual budget:



\$150000

How much of our budget we've secured to date:



\$50000

[Our promotional video:](#)



Our promotional materials:



<https://flatbox.co/waterbox/>

<http://www.prensalibre.com/economia/economia/flatbox-tecnologia-a-favor-del-agua>

<http://www.innovationworldcup.com/2017/01/iot-innovators-of-the-year-2017/>



Partnership Needs

SOLUTION TEAM



**EDWIN KESTLER**

CTO, Flatbox, S.A.

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